Emergency responder radio coverage test: Example: public safety ERCES, 3-storey office (completed example)

Example office building, Arlington VA (fictional) · Example County Fire Marshal (example) · IFC 2021 510 + NFPA 1225 (2022) · prepared 2026-10-28

Result: PASS

Packet index: NFPA 1225 (2022) and IFC 2021 510.5.4 (no AHJ list)

NFPA 1225-2022 A.18.2.1 and 20.3.10.1.3; IFC 2021 510.5.4 · oci.georgia.gov · every item is in the packet

Acceptance test report

#	Item	Section	Where it is
1	Equipment list and data sheets	NFPA 1225 20.3.10.1.3	Design submittal §4
2	As-built drawings	NFPA 1225 20.3.10.1.3	Attached: As-built drawings (example).pdf
3	Measurements at all locations covered	NFPA 1225 20.3.10.1.3	Report: floor pages
4	Secondary power calculation	NFPA 1225 20.3.10.1.3	Design submittal §5
5	Assigned frequencies	NFPA 1225 20.3.10.1.3	Report: Systems covered
6	Isolation test results	NFPA 1225 20.3.10.1.3	Report: Acceptance tests (pass)
7	Measured source levels and source type	NFPA 1225 20.3.10.1.3	Report: Acceptance tests (pass)
8	Gain of each channel or band	NFPA 1225 20.3.10.1.3; IFC 2021 510.5.4(6)	Report: Acceptance tests (pass)
9	Filter bandwidths and AGC parameters	NFPA 1225 20.3.10.1.3	Report: Acceptance tests (pass)
10	Spurious oscillation check	IFC 2021 510.5.4(7)	Report: Acceptance tests (pass)
11	Two-radio test (one within 10 ft of an antenna, one farthest)	IFC 2021 510.5.4(8)	Report: Acceptance tests (pass)
12	Test equipment and its calibration	IFC 2021 510.5.4(2)	Report: Tester and instruments

Design submittal

#	Item	Section	Where it is
1	DAQ signal source level measurements	NFPA 1225 A.18.2.1	Report: Systems covered
2	Statement of compliance with the local code	NFPA 1225 A.18.2.1	Report: Tester's statement
3	Site, floor and elevation plans	NFPA 1225 A.18.2.1	Design submittal §7
4	Donor path profiles and link budgets	NFPA 1225 A.18.2.1	Design submittal §2
5	Cable diagrams and device locations (riser)	NFPA 1225 A.18.2.1	Report: System diagram
6	Backup power calculations	NFPA 1225 A.18.2.1	Design submittal §5
7	Donor / DAS isolation calculations	NFPA 1225 A.18.2.1	Design submittal §2
8	Predictive DAQ or signal coverage maps	NFPA 1225 A.18.2.1(14)	Design submittal §3
9	Designer and installer qualifications	NFPA 1225 A.18.2.1	Report: Tester and instruments
10	Test grids on the floor plans	NFPA 1225 A.18.2.1	Design submittal §7
11	Cut sheets for all equipment	NFPA 1225 A.18.2.1	Attached: spec sheets

Pass criteria: each test location ≥ -95 dBm downlink and uplink, DAQ ≥ 3 both ways (or SINR ≥ 18 dB with BER ≤ 2.5%; RF Recon default: NFPA 1225 A.20.3.10, DAQ 3.0 = SINR 18 dB / BER 2.5%), on every channel tested there (the worst channel is shown); 95% of general areas and 99% of critical areas on each floor. Each floor's grid: IFC 510.5.4: 20 areas per floor; more than 1 failed area fails the floor. With exactly 2 failed, the floor may be re-divided into 40 areas and then passes with no more than 2 failed areas, none sharing an edge with another; failing that, the system is altered to reach 95%. Critical squares and named critical areas also meet 99% (profile: IFC 2021 (refs NFPA 1221): –95 dBm + DAQ 3.0). Only readings taken with an instrument calibrated within 12 months of the reading are scored.

Building and people

Building	Example: public safety ERCES, 3-storey office (completed example) · Example office building, Arlington VA (fictional) · 3 floors · 72,000 sq ft · Thermally efficient (low-E glass, foil insulation), typical (31 dB)
AHJ / fire marshal	Dana Whitfield · Example County Fire Marshal's Office (example) · 703-555-0142 · dwhitfield@firemarshal.example
Building owner	Priya Raman · Sample Properties LLC (example) · 703-555-0117 · praman@sampleproperties.example
Radio system owner	Marcus Bell · Example County ECC radio shop (example) · 703-555-0163 · radioshop@county-ecc.example
General contractor	Tom Alvarez · Sample Builders Inc. (example) · 703-555-0125 · talvarez@samplebuilders.example
Subcontractor	Lena Ortiz · Sample Electric Co. (example) · 703-555-0138 · lortiz@sampleelectric.example
Our technician	Jordan Lee · Sample RF Services (example) · 703-555-0109 · jlee@samplerf.example

Tester and instruments

Tester	Jordan Lee (example)
FCC GROL	PG-EX-0001 (example)
NICET	Level III (example)
Test dates	2026-10-20 to 2026-10-21

Instrument	Serial	Calibrated	Certificate	Status
PCTEL IBflex	EX-1001	2026-06-15	Calibration certificate - PCTEL IBflex EX-1001 (example).pdf, to 2027-06-15	current
Anritsu S412E LMR Master	EX-3001	2026-04-02	Calibration certificate - Anritsu S412E EX-3001 (example).pdf, to 2027-04-02	current

Compliance

Licensee / agency consent to retransmit (47 CFR 90.219(b)(1))	2026-09-02
FCC signal booster registration (Class B: 47 CFR 90.219(d)(5))	2026-10-22
AHJ permit	2026-08-15
Design approved	2026-09-10
Monitoring connected to the fire alarm panel	2026-10-16
Acceptance test passed	2026-10-27
Battery test	2026-10-21
Annual retest due	2027-10-27
Compared with	the acceptance test of 2026-10-27
Material rules confirmed	22: UL 2524 listing of active equipment; Enclosures for BDAs and batteries; Secondary (standby) power; Supervisory signals to the fire alarm system; Dedicated annunciator; Backbone survivability (fire resistance); Older-edition survivability rules (jurisdictions still on them); Cable fire ratings (plenum / riser); Mechanical protection of coax; Lightning / surge protection on the donor line; Grounding the coax shield at entry; Donor antenna mounting and isolation; Oscillation control; FCC Part 90 signal booster class, consent and registration; Frequencies and upgradeability; Near-far and antenna density; Link budget: choosing cable size; Component power ratings; Low-PIM choices for multi-carrier DAS; Radiating cable stop bands; Fiber connectors for RF-over-fiber DAS; Labelling and signage

Systems covered

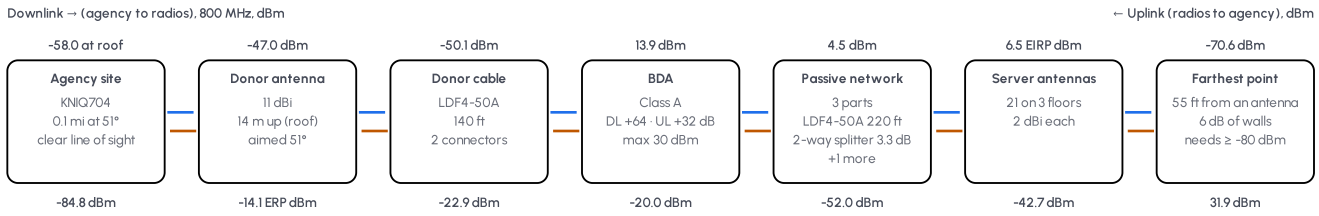
Agency	Call sign	Downlink MHz	Uplink MHz	Type
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	851.7125	806.7125	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	851.8125	806.8125	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	852.675	807.675	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	853.1875	808.1875	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	856.4375	811.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	856.7625	811.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	856.9375	811.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	857.4375	812.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	857.7625	812.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	857.9375	812.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	858.4375	813.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	858.7625	813.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	858.9375	813.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	859.4375	814.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	859.7625	814.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	859.9375	814.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	860.4375	815.4375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	860.7625	815.7625	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KNIQ704	860.9375	815.9375	Analog FM / Unknown
Arlington County Department of Public Safety Communications and Emergency Management	KIC338	154.13	153.89	Analog FM
Arlington County Department of Public Safety Communications and Emergency Management	KIC338	154.28	153.89	Analog FM

Licensee consent (47 CFR 90.219(b)(1))

Licensee	Call signs	Consent	Date
Arlington County Department of Public Safety Communications and Emergency Management	KIC338, KNIQ704	given	2026-09-02

Licensee	Call signs	Consent	Date
METROPOLITAN WASHINGTON AIRPORTS AUTHORITY	WPAY961, WPQB443, WQYK669	not needed	2026-09-02
City of Alexandria	KA9678, WNFS909, WQPF485, WQPG622	not needed	2026-09-02
MONTGOMERY, COUNTY OF	KCP601, WNPN638, WPBH837, WPWJ571, WPWL719, WREP612, WRHY973, WRHY978	not needed	2026-09-02
Radio Communications	WNHB839, WPVA690, WPVK244, WZX459	not needed	2026-09-02
Prince George's County Government-OHS	KGA361, WNBR219, WQJS954, WQLP960, WQSH369, WRAN914, WRAN915, WRAP365, WRPB615	not needed	2026-09-02
Baltimore Gas and Electric Company	KNDB915, KNDH602, WQAP338, WQCK465, WQLC435, WQLY679	not needed	2026-09-02
DISTRICT OF COLUMBIA	KNJA391, KNJE616, KNJU834, WPXT459, WPYD604, WRBZ373	not needed	2026-09-02
WASHINGTON GAS LIGHT COMPANY	WRVQ768, WRWK328, WRWR905, WRXC754, WRXC755, WRXC757	not needed	2026-09-02
FAIRFAX, COUNTY OF	KNIH412, WNAJ365, WNYZ445, WNYZ447, WPPA560, WPVY861, WQCP394	not needed	2026-09-02
MONTGOMERY COUNTY, MARYLAND	WQZC785, WQZC983	not needed	2026-09-02
Fairfax County Public Schools	WRJI389	not needed	2026-09-02
State of Maryland - UNIVERSITY OF MARYLAND	WPTZ482	not needed	2026-09-02
Howard, County of MD	WPXJ396	not needed	2026-09-02

System diagram: signal path, start to finish



Levels are the 800 MHz link budget's worst case (860.9375 MHz down, 815.9375 MHz up); every band is in the table below. Blue: what reaches the radios inside; orange: what reaches the agency's receiver.

Band	Systems	At roof dBm	BDA out dBm	Antenna EIRP dBm	Farthest point dBm	Needs dBm	Uplink at site dBm	Checks
800 MHz	Arlington County Department of Public Safety Communications and Emergency Management	-58.0	13.9	6.5	-70.6	-80.0	-84.8	pass
VHF High	Arlington County Department of Public Safety Communications and Emergency Management	-60.0	15.6	8.9	-51.2	-80.0	-82.9	pass

Donor path to	Distance mi	Bearing	Terrain	Terrain loss dB
KNIQ704	0.08	51°	clear line of sight, 7.4 of the first Fresnel zone clear	0
KIC338	0.58	238°	clear line of sight, 0.78 of the first Fresnel zone clear	0

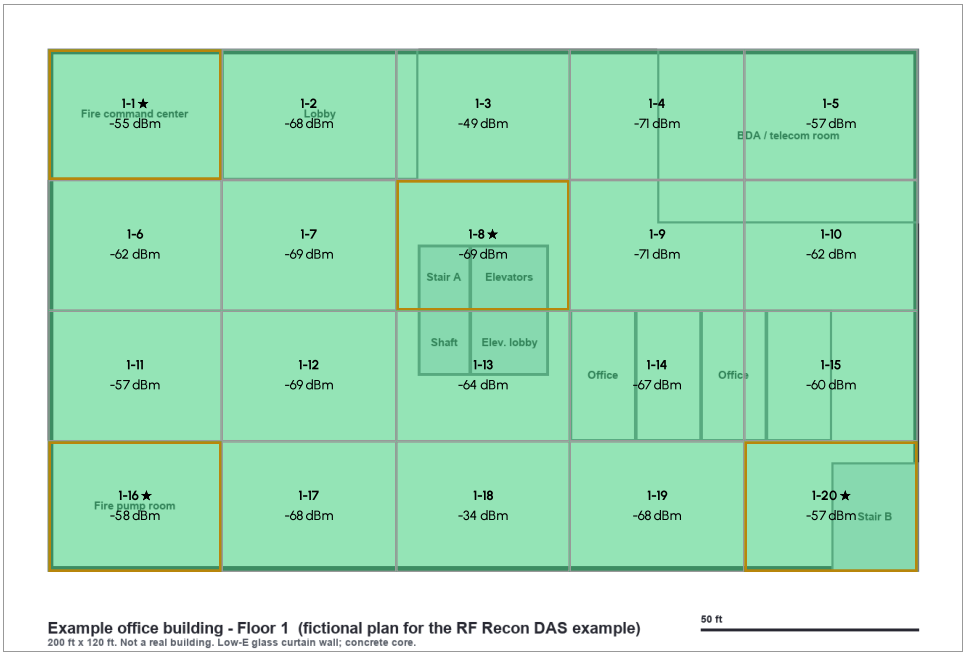
Antennas by floor: Floor 1: 7 · Floor 2: 7 · Floor 3: 7

Summary by floor

Floor	General pass	Critical pass	Untested	Not counted	Result	Rule and reason
1	16/16 (100%)	4/4 (100%)	0		pass	IFC 510.5.4, 20 areas: No area failed.
2	18/18 (100%)	2/2 (100%)	0		pass	IFC 510.5.4, 20 areas: No area failed.
3	18/18 (100%)	2/2 (100%)	0		pass	IFC 510.5.4, 20 areas: No area failed.
Named critical areas		5/5	0			each must pass

Floor 1: 16/16 general, 4/4 critical pass · PASS

IFC 510.5.4, 20 areas: No area failed.

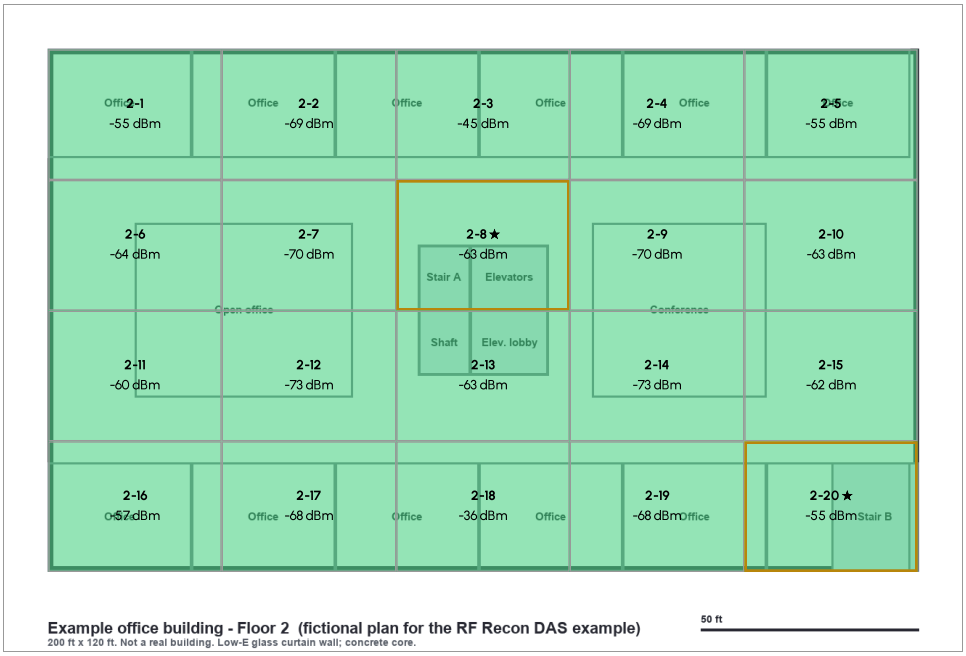


Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
1-1	★	154.13 (worst of 2)	−55	4.6		−62	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
1-2		154.13 (worst of 2)	−68	4.4		−79	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
1-3		851.7125 (worst of 2)	−49	4.6		−59	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
1-4		851.7125 (worst of 2)	−71	4.2		−76	4	+0 dB	PCTEL IBflex EX-1001	pass	
1-5		154.13 (worst of 2)	−57	4.4		−67	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
1-6		851.7125 (worst of 2)	−62	4.2		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-7		851.7125 (worst of 2)	−69	4.1		−73	3.8	+0 dB	PCTEL IBflex EX-1001	pass	
1-8	★	851.7125 (worst of 2)	−69	4.4		−81	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
1-9		851.7125 (worst of 2)	−71	4.2		−83	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
1-10		851.7125 (worst of 2)	−62	4.3		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-11		154.13 (worst of 2)	−57	4.3		−66	3.9	+0 dB	PCTEL IBflex EX-1001	pass	
1-12		851.7125 (worst of 2)	−69	4.2		−80	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-13		154.13 (worst of 2)	−64	4.4		−73	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
1-14		154.13 (worst of 2)	−67	4.1		−81	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
1-15		154.13 (worst of 2)	−60	4.3		−70	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
1-16	★	851.7125 (worst of 2)	−58	4.4		−68	4.1	+0 dB	PCTEL IBflex EX-1001	pass	Fire pump room: pump running, door closed; the fire marshal witnessed this spot (example).

Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
1-17		851.7125 (worst of 2)	−68	4.6		−77	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
1-18		851.7125 (worst of 2)	−34	4.6		−42	4.7	+0 dB	PCTEL IBflex EX-1001	pass	
1-19		154.13 (worst of 2)	−68	4.4		−78	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
1-20	★	851.7125 (worst of 2)	−57	4.6		−66	4.5	+0 dB	PCTEL IBflex EX-1001	pass	

Floor 2: 18/18 general, 2/2 critical pass · PASS

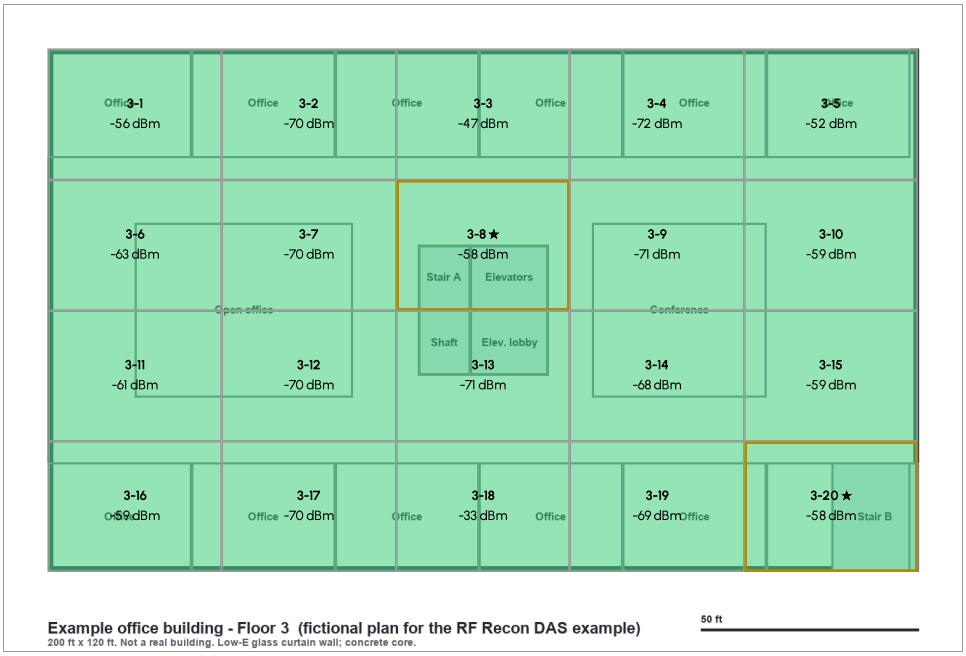
IFC 510.5.4, 20 areas: No area failed.



Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
2-1		154.13 (worst of 2)	−55	4.6		−66	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
2-2		851.7125 (worst of 2)	−69	4.4		−75	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
2-3		851.7125 (worst of 2)	−45	4.5		−55	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
2-4		154.13 (worst of 2)	−69	4.4		−75	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-5		851.7125 (worst of 2)	−55	4.5		−58	4.5	+0 dB	PCTEL IBflex EX-1001	pass	
2-6		154.13 (worst of 2)	−64	4.3		−72	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-7		154.13 (worst of 2)	−70	4.3		−80	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-8	★	154.13 (worst of 2)	−63	4.5		−80	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
2-9		851.7125 (worst of 2)	−70	4.2		−73	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
2-10		851.7125 (worst of 2)	−63	4.7		−71	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
2-11		851.7125 (worst of 2)	−60	4.4		−70	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
2-12		851.7125 (worst of 2)	−73	4.1		−85	4	+0 dB	PCTEL IBflex EX-1001	pass	
2-13		851.7125 (worst of 2)	−63	4.5		−68	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
2-14		154.13 (worst of 2)	−73	4.3		−81	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
2-15		154.13 (worst of 2)	−62	4.5		−69	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
2-16		851.7125 (worst of 2)	−57	4.4		−69	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
2-17		851.7125 (worst of 2)	−68	4.4		−76	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
2-18		154.13 (worst of 2)	−36	4.5		−48	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
2-19		154.13 (worst of 2)	−68	4.4		−76	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
2-20	★	851.7125 (worst of 2)	−55	4.5		−67	4.3	+0 dB	PCTEL IBflex EX-1001	pass	

Floor 3: 18/18 general, 2/2 critical pass · PASS

IFC 510.5.4, 20 areas: No area failed.



Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
3-1		851.7125 (worst of 2)	-56	4.7		-65	4.6	+0 dB	PCTEL IBflex EX-1001	pass	
3-2		851.7125 (worst of 2)	-70	4.3		-79	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-3		851.7125 (worst of 2)	-47	4.7		-55	4.5	+0 dB	PCTEL IBflex EX-1001	pass	
3-4		851.7125 (worst of 2)	-72	4.1		-82	3.8	+0 dB	PCTEL IBflex EX-1001	pass	
3-5		154.13 (worst of 2)	-52	4.5		-59	4.4	+0 dB	PCTEL IBflex EX-1001	pass	Rechecked after the floor 3 remote's VHF gain was raised 3 dB on 2026-10-17 (it read −94.5 dBm at the pre-acceptance check).
3-6		851.7125 (worst of 2)	-63	4.4		-66	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-7		851.7125 (worst of 2)	-70	4.4		-80	4.4	+0 dB	PCTEL IBflex EX-1001	pass	
3-8	★	851.7125 (worst of 2)	-58	4.4		-69	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-9		154.13 (worst of 2)	-71	4.3		-80	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-10		851.7125 (worst of 2)	-59	4.4		-64	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
3-11		851.7125 (worst of 2)	-61	4.4		-67	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-12		851.7125 (worst of 2)	-70	4.4		-76	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-13		851.7125 (worst of 2)	-71	4.3		-77	4.1	+0 dB	PCTEL IBflex EX-1001	pass	
3-14		154.13 (worst of 2)	-68	4.3		-74	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
3-15		851.7125 (worst of 2)	-59	4.5		-68	4.5	+0 dB	PCTEL IBflex EX-1001	pass	
3-16		851.7125 (worst of 2)	-59	4.6		-70	4.5	+0 dB	PCTEL IBflex EX-1001	pass	

Grid	Critical	Channel	DL dBm	DL DAQ	SINR / BER	UL dBm	UL DAQ	vs baseline	Instrument	Result	Note
3-17		154.13 (worst of 2)	−70	4.1		−78	4	+0 dB	PCTEL IBflex EX-1001	pass	
3-18		154.13 (worst of 2)	−33	4.6		−41	4.3	+0 dB	PCTEL IBflex EX-1001	pass	
3-19		851.7125 (worst of 2)	−69	4.3		−71	4.2	+0 dB	PCTEL IBflex EX-1001	pass	
3-20	★	154.13 (worst of 2)	−58	4.5		−65	4.2	+0 dB	PCTEL IBflex EX-1001	pass	

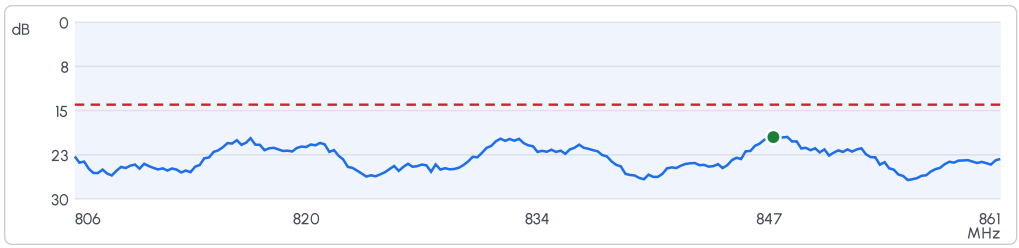
Named critical areas

Area	DL dBm	DL DAQ	UL dBm	UL DAQ	Instrument	Result
Fire command center	−71	4.4	−75	4.1	PCTEL IBflex EX-1001	pass
Fire pump room	−84	3.8	−86	3.6	PCTEL IBflex EX-1001	pass
Exit stairs	−82	3.9	−84	3.7	PCTEL IBflex EX-1001	pass
Elevator lobbies	−77	4.2	−80	4	PCTEL IBflex EX-1001	pass
BDA / telecom room	−69	4.5	−72	4.3	PCTEL IBflex EX-1001	pass

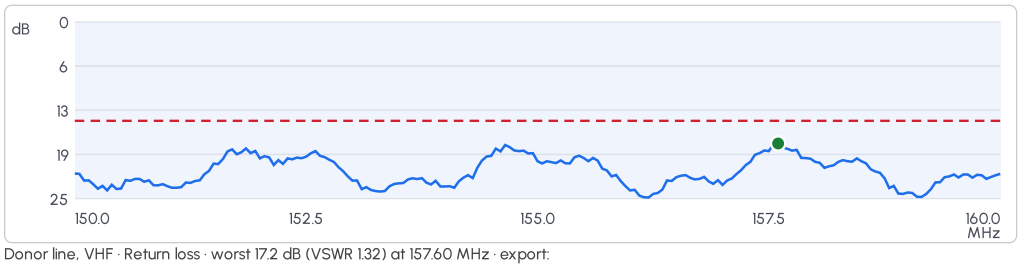
PIM and sweeps

NFPA 1225 (2022) chapter 18 asks for no PIM test and no sweep (https://oci.georgia.gov/document/document/nfpa-1225-chapter-18/download); these were run because the AHJ, the owner or a carrier asked for them.

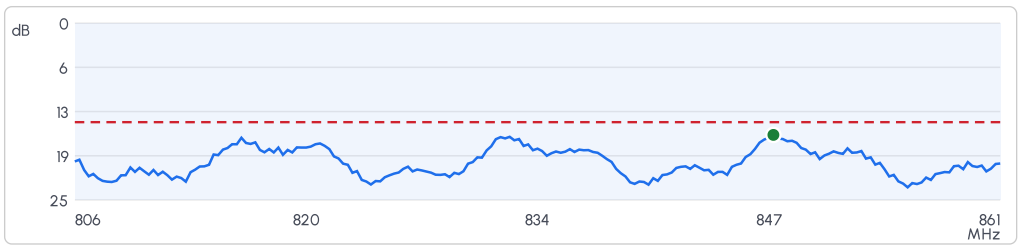
Sweep	Measurement	Instrument	Worst	Where on the trace	Limit and source	Result
Donor line, 800 MHz	Return loss	Anritsu S412E EX-3001 · 2026-10-15	**19.5 dB RL**	847.52 MHz	return loss ≥ 14 dB (VSWR ≤ 1.5:1): VSWR 1.5:1 (14 dB return loss) (The VSWR 1.5:1 rating on public-safety coax (e.g. the 2-hour rated coax in the parts research): common practice, not a code rule.)	pass
Donor line, VHF	Return loss	Anritsu S412E EX-3001 · 2026-10-15	**17.2 dB RL**	157.60 MHz	return loss ≥ 14 dB (VSWR ≤ 1.5:1): VSWR 1.5:1 (14 dB return loss) (The VSWR 1.5:1 rating on public-safety coax (e.g. the 2-hour rated coax in the parts research): common practice, not a code rule.)	pass
Riser to floor 3, 800 MHz	Return loss	Anritsu S412E EX-3001 · 2026-10-15	**15.8 dB RL**	847.52 MHz	return loss ≥ 14 dB (VSWR ≤ 1.5:1): VSWR 1.5:1 (14 dB return loss) (The VSWR 1.5:1 rating on public-safety coax (e.g. the 2-hour rated coax in the parts research): common practice, not a code rule.)	pass



Donor line, 800 MHz · Return loss · worst 19.5 dB (VSWR 1.24) at 847.52 MHz · export:



Donor line, VHF · Return loss · worst 17.2 dB (VSWR 1.32) at 157.60 MHz · export:



Riser to floor 3, 800 MHz · Return loss · worst 15.8 dB (VSWR 1.39) at 847.52 MHz · export:

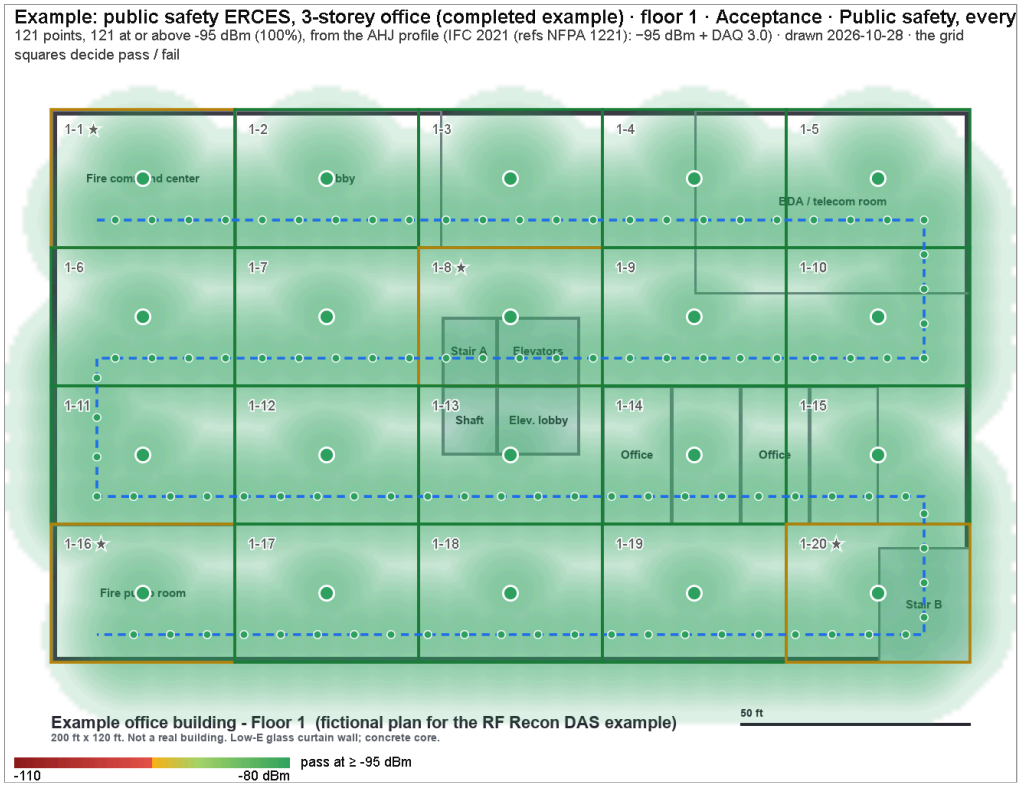
PIM runs

Where	Band	F1	F2	Tone dBm	Order	PIM dBm	dBc	Result (≤ -140 dBc)
Head-end, BDA output	800 PS	**851.7125**	**856.4375**	**43**	IM3	**-104**	**-147.0**	pass
Riser to floor 3	800 PS	**851.7125**	**856.4375**	**43**	IM3	**-101**	**-144.0**	pass
Floor 2 branch	800 PS	**851.7125**	**856.4375**	**43**	IM3	**-92**	**-135.0**	fail
Floor 2 branch (retest)	800 PS	**851.7125**	**856.4375**	**43**	IM3	**-106**	**-149.0**	pass

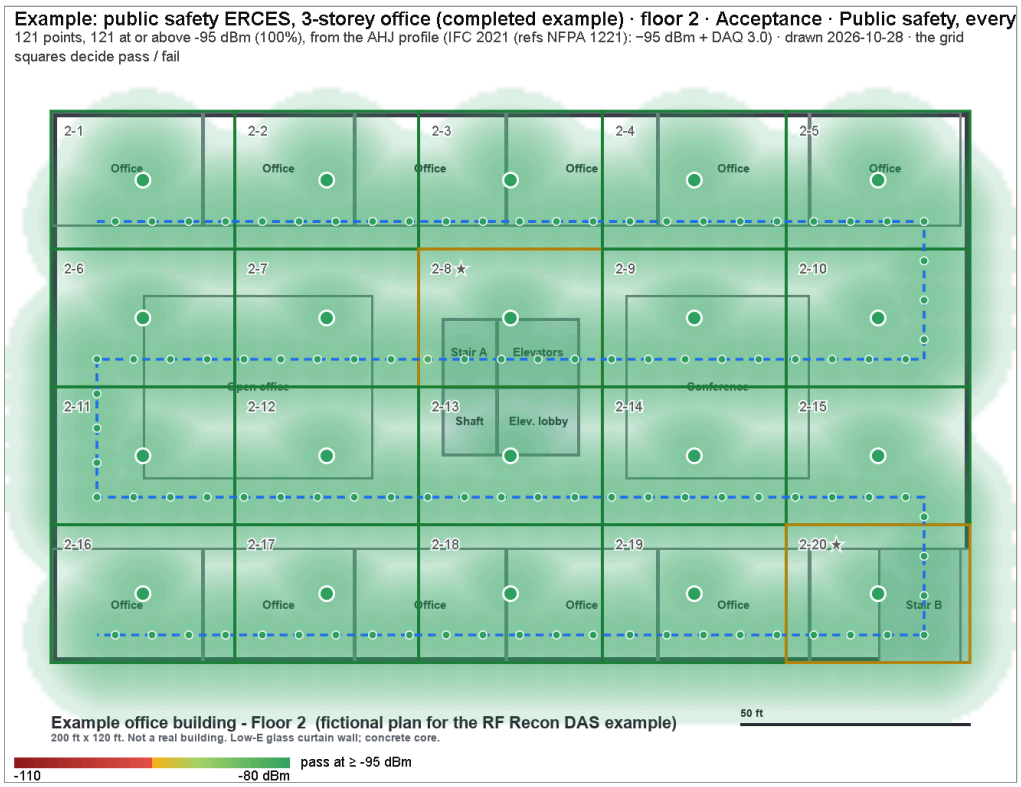
PIM limit -140 dBc: the job's own (PIM tab); –140 dBc at 2 × 43 dBm is the common system-level pass in the PIM research, not a code rule.

Walk test heat maps: Acceptance

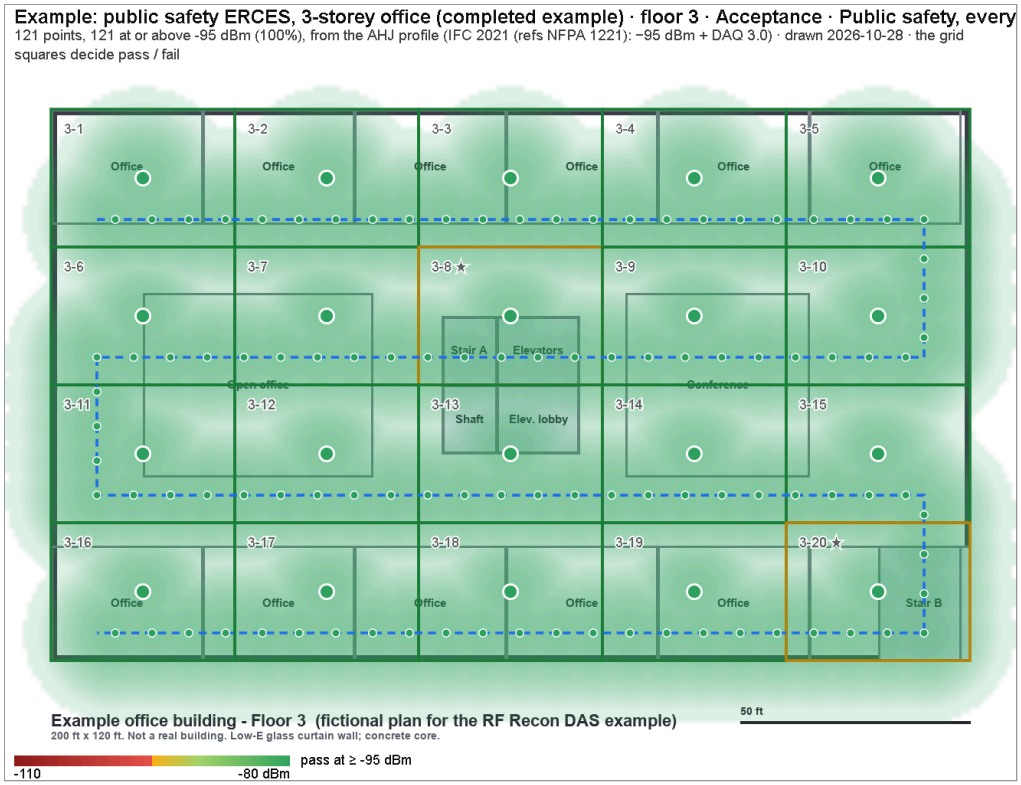
Readings walked on each floor's plan, coloured against the requirement. The heat map shows every reading, calibrated or not; the grid squares above decide pass / fail.



Floor 1 · Acceptance · Public safety, every channel (worst) · DL dBm · 121 points, 121 at or above -95 dBm (100%), from the AHJ profile (IFC 2021 (refs NFPA 1221): −95 dBm + DAQ 3.0) · drawn 2026-10-28 · the grid squares decide pass / fail



Floor 2 · Acceptance · Public safety, every channel (worst) · DL dBm · 121 points, 121 at or above -95 dBm (100%), from the AHJ profile (IFC 2021 (refs NFPA 1221): −95 dBm + DAQ 3.0) · drawn 2026-10-28 · the grid squares decide pass / fail

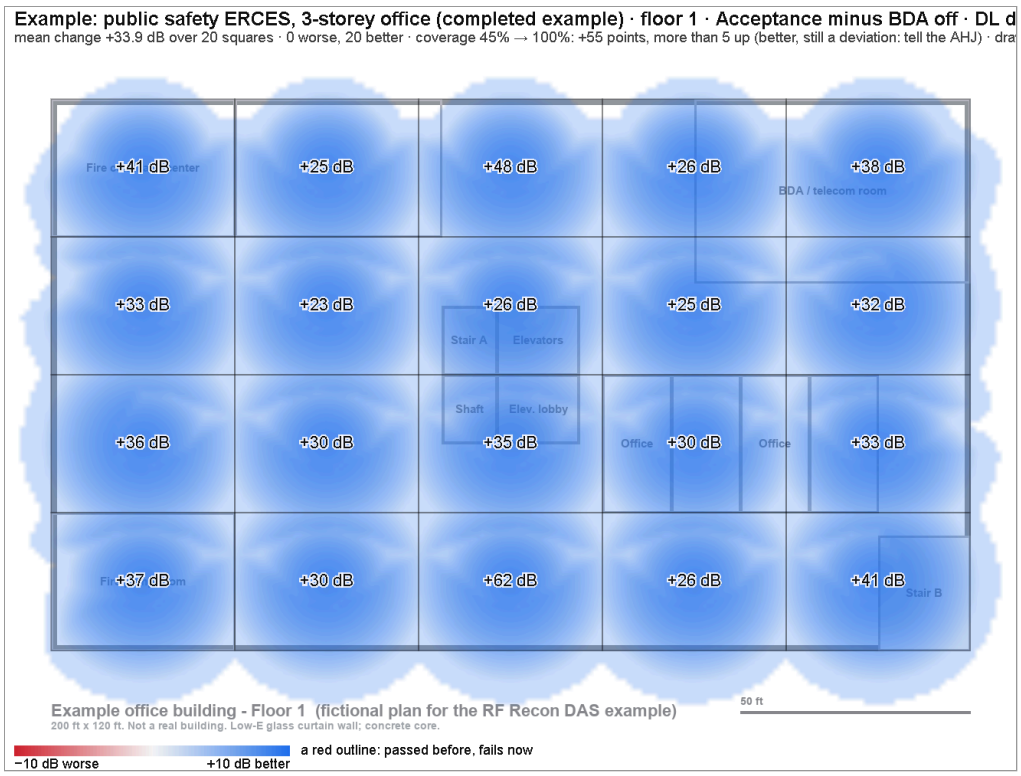


Floor 3 · Acceptance · Public safety, every channel (worst) · DL dBm · 121 points, 121 at or above -95 dBm (100%), from the AHJ profile (IFC 2021 (refs NFPA 1221): −95 dBm + DAQ 3.0) · drawn 2026-10-28 · the grid squares decide pass / fail

Test passes

Pass	Points	Squares passing, by floor (public safety, scored)	
BDA off	**60**	1: 45% of 20 · 2: 0% of 20 · 3: 0% of 20	
BDA on	**60**	1: 100% of 20 · 2: 100% of 20 · 3: 100% of 20	
Acceptance	**363**	1: 100% of 20 · 2: 100% of 20 · 3: 100% of 20	the grid above

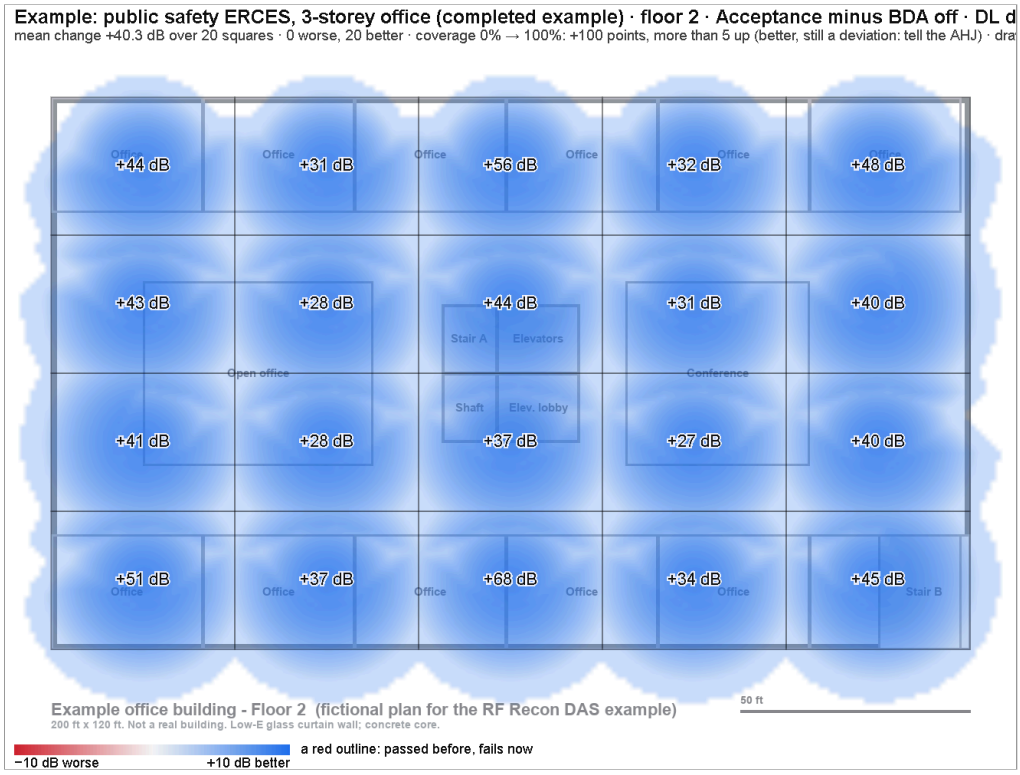
Acceptance compared with BDA off



Floor 1 · Acceptance minus BDA off · DL dBm · coverage 45% → 100%: +55 points, more than 5 up (better, still a deviation: tell the AHJ)

Outside (better) Clark County's 5%: coverage 45% → 100%: +55 points, more than 5 up (better, still a deviation: tell the AHJ) (20 and 121 points). Clark County FPAN, five-year: no deviation of coverage more than 5 percent from the initial installation documentation (FPAN Guide, Dec 2025). Coverage: the tested squares that pass, public safety, by the job's AHJ profile.

Square	BDA off	Acceptance	Change	Result then → now
1-1	−96 dBm	−55 dBm	+41	fail → pass
1-2	−93 dBm	−68 dBm	+25	pass → pass
1-3	−97 dBm	−49 dBm	+48	fail → pass
1-4	−97 dBm	−71 dBm	+26	fail → pass
1-5	−95 dBm	−57 dBm	+38	pass → pass
1-6	−95 dBm	−62 dBm	+33	fail → pass
1-7	−92 dBm	−69 dBm	+23	pass → pass
1-8	−95 dBm	−69 dBm	+26	pass → pass
1-9	−96 dBm	−71 dBm	+25	fail → pass
1-10	−94 dBm	−62 dBm	+32	pass → pass
1-11	−93 dBm	−57 dBm	+36	pass → pass
1-12	−99 dBm	−69 dBm	+30	fail → pass
1-13	−99 dBm	−64 dBm	+35	fail → pass
1-14	−97 dBm	−67 dBm	+30	fail → pass
1-15	−93 dBm	−60 dBm	+33	pass → pass
1-16	−95 dBm	−58 dBm	+37	pass → pass
1-17	−98 dBm	−68 dBm	+30	fail → pass
1-18	−96 dBm	−34 dBm	+62	fail → pass
1-19	−94 dBm	−68 dBm	+26	pass → pass
1-20	−98 dBm	−57 dBm	+41	fail → pass

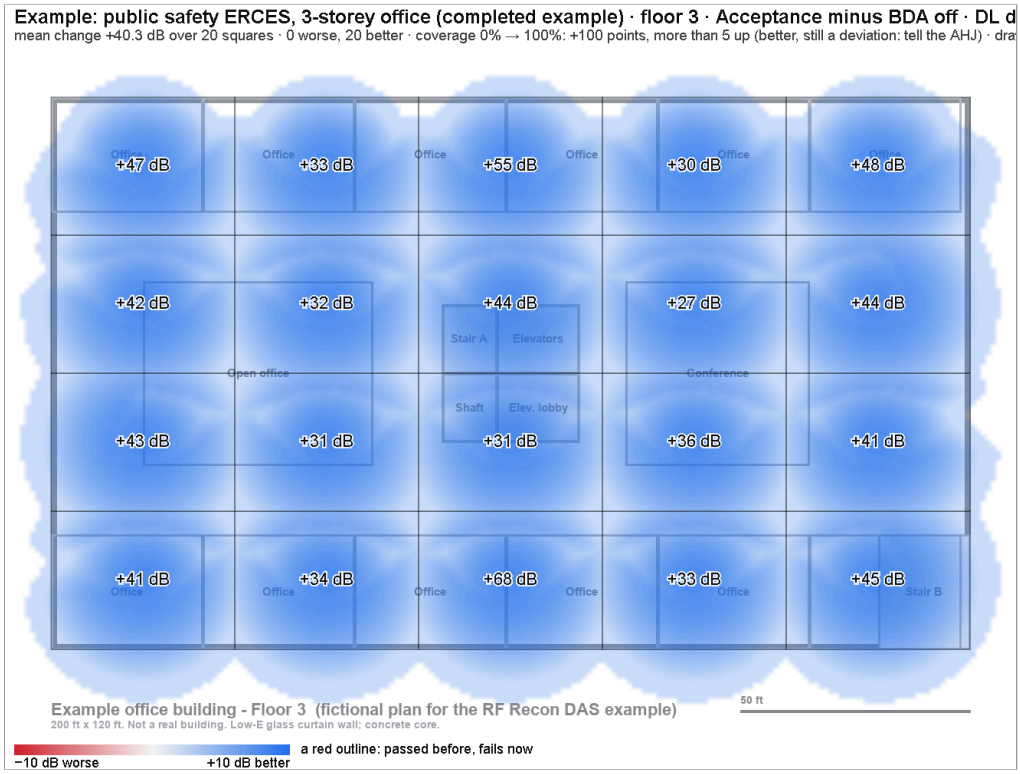


Floor 2 · Acceptance minus BDA off · DL dBm · coverage 0% → 100%: +100 points, more than 5 up (better, still a deviation: tell the AHJ)

Outside (better) Clark County's 5%: coverage 0% → 100%: +100 points, more than 5 up (better, still a deviation: tell the AHJ) (20 and 121 points). Clark County FPAN, five-year: no deviation of coverage more than 5 percent from the initial installation documentation (FPAN Guide, Dec 2025). Coverage: the tested squares that pass, public safety, by the job's AHJ profile.

Square	BDA off	Acceptance	Change	Result then → now
2-1	−99 dBm	−55 dBm	+44	fail → pass
2-2	−100 dBm	−69 dBm	+31	fail → pass

Square	BDA off	Acceptance	Change	Result then → now
2-3	-101 dBm	-45 dBm	+56	fail → pass
2-4	-101 dBm	-69 dBm	+32	fail → pass
2-5	-103 dBm	-55 dBm	+48	fail → pass
2-6	-107 dBm	-64 dBm	+43	fail → pass
2-7	-98 dBm	-70 dBm	+28	fail → pass
2-8	-107 dBm	-63 dBm	+44	fail → pass
2-9	-101 dBm	-70 dBm	+31	fail → pass
2-10	-103 dBm	-63 dBm	+40	fail → pass
2-11	-101 dBm	-60 dBm	+41	fail → pass
2-12	-101 dBm	-73 dBm	+28	fail → pass
2-13	-100 dBm	-63 dBm	+37	fail → pass
2-14	-100 dBm	-73 dBm	+27	fail → pass
2-15	-102 dBm	-62 dBm	+40	fail → pass
2-16	-108 dBm	-57 dBm	+51	fail → pass
2-17	-105 dBm	-68 dBm	+37	fail → pass
2-18	-104 dBm	-36 dBm	+68	fail → pass
2-19	-102 dBm	-68 dBm	+34	fail → pass
2-20	-100 dBm	-55 dBm	+45	fail → pass



Floor 3 · Acceptance minus BDA off · DL dBm · coverage 0% → 100%: +100 points, more than 5 up (better, still a deviation: tell the AHJ)

Outside (better) Clark County's 5%: coverage 0% → 100%: +100 points, more than 5 up (better, still a deviation: tell the AHJ) (20 and 121 points). Clark County FPAN, five-year: no deviation of coverage more than 5 percent from the initial installation documentation (FPAN Guide, Dec 2025). Coverage: the tested squares that pass, public safety, by the job's AHJ profile.

Square	BDA off	Acceptance	Change	Result then → now
3-1	-103 dBm	-56 dBm	+47	fail → pass
3-2	-103 dBm	-70 dBm	+33	fail → pass
3-3	-102 dBm	-47 dBm	+55	fail → pass
3-4	-102 dBm	-72 dBm	+30	fail → pass
3-5	-100 dBm	-52 dBm	+48	fail → pass

Square	BDA off	Acceptance	Change	Result then → now
3-6	−105 dBm	−63 dBm	+42	fail → pass
3-7	−102 dBm	−70 dBm	+32	fail → pass
3-8	−102 dBm	−58 dBm	+44	fail → pass
3-9	−98 dBm	−71 dBm	+27	fail → pass
3-10	−103 dBm	−59 dBm	+44	fail → pass
3-11	−104 dBm	−61 dBm	+43	fail → pass
3-12	−101 dBm	−70 dBm	+31	fail → pass
3-13	−102 dBm	−71 dBm	+31	fail → pass
3-14	−104 dBm	−68 dBm	+36	fail → pass
3-15	−100 dBm	−59 dBm	+41	fail → pass
3-16	−100 dBm	−59 dBm	+41	fail → pass
3-17	−104 dBm	−70 dBm	+34	fail → pass
3-18	−101 dBm	−33 dBm	+68	fail → pass
3-19	−102 dBm	−69 dBm	+33	fail → pass
3-20	−103 dBm	−58 dBm	+45	fail → pass

Acceptance tests · 12 of 12 recorded · IFC 2021 510.5.4 (6)-(8); NFPA 1225-2022 20.3.10.1.3 (8)-(11)

Test	Result	Value	Note	By, on	Code
Amplifier gain, each channel or band	pass	2 bands recorded	Sample Wireless (example) EX-BDA-2B serial EXB-24117 FCC ID EXAMPLE-BDA2B firmware 3.2.1	Jordan Lee (example), 2026-10-16	IFC 2021 510.5.4(6); NFPA 1225 20.3.10.1.3
Filter bandwidths and AGC settings	pass	12.5 kHz per channel; AGC on both bands	read from the BDA, screenshot in the commissioning record	Jordan Lee (example), 2026-10-16	NFPA 1225 20.3.10.1.3; Clark County FPAN §7.0 item 10
Donor-to-server antenna isolation, measured	pass	34 dB over gain	800 MHz: 108 dB vs 84 dB needed (gain 64 + 20); VHF High: 104 dB vs 90 dB needed (gain 70 + 20) · margin 20 dB from NFPA 1225 (2022) and IFC 2021	Jordan Lee (example), 2026-10-16	NFPA 1225 18.10, 20.3.10.1.3; San Diego C-1 App. D §VI.G
Noise floor: ambient, active and quiescent	pass	0.4 dB rise	3 of 3 channels under 1 dB · KNIQ704 master site, 0.08 mi with Marcus Bell, county radio shop (example)	Jordan Lee (example), 2026-10-16	NFPA 1225 A.18.9.3; San Diego C-1 App. D §VI.D-F
Spurious oscillation check (spectrum analyzer)	pass	none seen	oscillation alarm and shutdown seen working · with Anritsu S412E EX-3001 · gain raised until the BDA saw oscillation and cut back, then set back to the values above	Jordan Lee (example), 2026-10-16	IFC 2021 510.5.4(7)
Two-radio test: one within 10 ft of an indoor antenna, one at the farthest point	pass	4 of 4 calls clear	County dispatch (example), County radio shop (example)	Jordan Lee (example), 2026-10-27	IFC 2021 510.5.4(8)

Test	Result	Value	Note	By, on	Code
Battery load test, 1 hour	pass	1 h	full load for 1 h on battery; 25.1 V at the end; the 70% low-battery signal reached the panel in the 12 h test	Jordan Lee (example), 2026-10-21	IFC 2021 510.6.1(3); San Diego C-1 App. D §VI.B
Supervisory signals to the fire alarm panel and annunciator	pass	8 of 8	each NFPA 1225 18.14.1.2 signal shown at the fire alarm panel and the annunciator	Jordan Lee (example), 2026-10-16	NFPA 1225 18.14; IFC 2021 510.4.2.5
Emergency power off (EPO)	pass		the BDA room EPO shuts the BDA off; it restarts on reset	Jordan Lee (example), 2026-10-16	San Diego C-1 App. D §V
Exterior RF leakage below the donor signal	pass	22 dB	four points 100 ft outside: −69 to −72 dBm against −47 dBm at the donor	Jordan Lee (example), 2026-10-21	San Diego C-1 §VI.H.4 (15 dB); LA County V7-C3-S21 §IV.B.7.c (20 dB)
Donor signal level and source type	pass	−53.1 dBm	Off-air donor antenna · recorded; NFPA 1225 (2022) and IFC 2021 sets no window	Jordan Lee (example), 2026-10-16	NFPA 1225 20.3.10.1.3
BDA / headend / BBU: model, serial, FCC ID, class	pass	Sample Wireless EX-BDA-2B, serial EXB-24117, FCC ID EXAMPLE-BDA2B, Class A	battery EX-BBU-12H serial EXU-0912; fiber head end EX-FDAS-HE serial EXH-3318	Jordan Lee (example), 2026-10-16	San Diego C-1 App. D §IV; NFPA 1225 20.3.10.1.3

Tester's statement

I tested this emergency responder radio coverage system on 2026-10-20 to 2026-10-21 with the calibrated instruments listed in this report, to IFC 2021 510 + NFPA 1225 (2022) as Example County Fire Marshal (example) applies them. The results are the readings taken; none were changed.

Signature	Name, Sample RF Services (example)	Date
	Jordan Lee, RF technician, FCC GROL	2026-10-28

This product uses the FCC Data API but is not endorsed or certified by the FCC.

Sealed 2026-10-28 · verify code EX26-1028 · SHA-256 bd3e94c09ab7b542. Sealed copy of this report kept with the job; a changed job no longer matches it.